Physical Chemistry Cume **Oct 20 2017**

1. For the following gas-phase reaction

$$A_{(g)} + B_{(g)} \rightleftharpoons 2C_{(g)} + 3D_{(g)}$$

15.0mmol of A and 18.0 mmol of B were placed in a container at 600K and left to reach equilibrium. The total pressure of the system after equilibrium is reached is 1085 torr, and contains 10.0 mmol of C.

a) What is K_p^0 and $\Delta_r G^0$ at 600K, assuming they are all ideal gases.

b) If 8mmol of A, 12 mmol of B, 7 mmol of C and 10 mmol of D were placed in a container at 600K a total pressure of 760 torr, would the system be in equilibrium? If not, would the system move towards more reactants or more products?

2. The equilibrium constants for the following reaction are 0.050 at 300K and 0.073 at 500K.

$$2C_3H_{6\,(g)} \rightleftharpoons C_2H_{4\,(g)} + C_4H_{8\,(g)}$$

Obtain values for $\Delta_r G^0$ and $\Delta_r H^0$ for the reaction at 400K. Assume ΔH is independent of temperature between 300 and 500K.

3. For the following mechanism

$$B \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} E \xrightarrow{k_2} F$$

$$E \xrightarrow{k_3} X$$

Assuming a steady state condition applies, what is the observed rate constant, k_{obs}, for the rate of formation of F in terms of the elementary rate constants?